

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Borane-tetrahydrofuran complex, 1.5M**  
Cat No. : **R21730**  
Molecular Formula **C4 H11 BO**  
REACH registration number **-**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

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Flammable liquids  
Substances/mixtures which, in contact with water, emit flammable gases

Category 2 (H225)  
Category 1 (H260)

## Health hazards

Acute oral toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Carcinogenicity  
Specific target organ toxicity - (single exposure)

Category 4 (H302)  
Category 2 (H315)  
Category 1 (H318)  
Category 2 (H351)  
Category 3 (H335) (H336)

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

## Hazard Statements

H225 - Highly flammable liquid and vapor  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
EUH014 - Reacts violently with water  
EUH019 - May form explosive peroxides

## Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician  
P335 + P334 - Brush off loose particles from skin. Immerse in cool water/wrap in wet bandages

## 2.3. Other hazards

Toxic to terrestrial vertebrates

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This product does not contain any known or suspected endocrine disruptors

## Section 3: Composition/information on ingredients

### 3.2. Mixtures

| Component                                 | CAS No     | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                            |
|-------------------------------------------|------------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran                           | 109-99-9   | 203-726-8         | 90       | Flam. Liq. 2 (H225)<br>Acute Tox. 4 (H302)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)<br>STOT SE 3 (H336)<br>Carc. 2 (H351)<br>(EUH019)                      |
| Boron, trihydro(tetrahydrofuran)-, (T-4)- | 14044-65-6 | EEC No. 237-881-8 | 10       | Water-react. 1 (H260)<br>Acute Tox. 4 (H302)<br>Skin Irrit. 2 (H315)<br>Eye Dam. 1 (H318)<br>STOT SE 3 (H335)<br>Flam. Liq. 2 (H225)<br>(EUH014)<br>(EUH019) |

| Component       | Specific concentration limits (SCL's)                                    | M-Factor | Component notes |
|-----------------|--------------------------------------------------------------------------|----------|-----------------|
| Tetrahydrofuran | Acute Tox. 4 :: C>82.5%<br>Eye Irrit. 2 :: C>=25%<br>STOT SE 3 :: C>=25% | -        | -               |

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes eye burns. Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

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## 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Dry sand. Carbon dioxide (CO<sub>2</sub>). Powder. Do not use water or foam. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

Water.

### 5.2. Special hazards arising from the substance or mixture

Reacts violently with water. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Oxides of boron, Hydrogen.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Do not allow contact with water. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

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## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep refrigerated. Keep away from water or moist air. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 4.3

**Switzerland - Storage of hazardous substances**

SC 4.3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti> Storage class -

## 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component       | European Union                                                                                                              | The United Kingdom                                                                                                            | France                                                                                                                                                                                                                         | Belgium                                                                                                                               | Spain                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm (8h)<br>TWA: 150 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 300 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br><br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>Skin | TWA / VME: 50 ppm (8 heures). restrictive limit<br>TWA / VME: 150 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 100 ppm. restrictive limit<br>STEL / VLCT: 300 mg/m <sup>3</sup> . restrictive limit<br>Peau | TWA: 50 ppm 8 uren<br>TWA: 150 mg/m <sup>3</sup> 8 uren<br>STEL: 100 ppm 15 minuten<br>STEL: 300 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 100 ppm (15 minutos).<br>STEL / VLA-EC: 300 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 50 ppm (8 horas)<br>TWA / VLA-ED: 150 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component       | Italy                                                                                                                                                                                                          | Germany                                                                                                                                                                                                                                                    | Portugal                                                                                                                                | The Netherlands                                                                                                                        | Finland                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm 8 ore.<br>Time Weighted Average<br>TWA: 150 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>STEL: 100 ppm 15 minuti. Short-term<br>STEL: 300 mg/m <sup>3</sup> 15 minuti. Short-term<br>Pelle | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 150 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 20 ppm (8 Stunden). MAK<br>TWA: 60 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 40 ppm<br>Höhepunkt: 120 mg/m <sup>3</sup> | STEL: 100 ppm 15 minutos<br>STEL: 300 mg/m <sup>3</sup> 15 minutos<br>TWA: 50 ppm 8 horas<br>TWA: 150 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>STEL: 200 ppm 15 minuten<br>STEL: 600 mg/m <sup>3</sup> 15 minuten<br>TWA: 100 ppm 8 uren<br>TWA: 300 mg/m <sup>3</sup> 8 uren | TWA: 50 ppm 8 tunteina<br>TWA: 150 mg/m <sup>3</sup> 8 tunteina<br>STEL: 100 ppm 15 minuutteina<br>STEL: 300 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

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|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                                                                                                                                                         | Haut                                                                                                                                           |                                                                                                                                                              |                                                                                         |                                                                                                                                                                                           |
| <b>Component</b> | <b>Austria</b>                                                                                                                                                          | <b>Denmark</b>                                                                                                                                 | <b>Switzerland</b>                                                                                                                                           | <b>Poland</b>                                                                           | <b>Norway</b>                                                                                                                                                                             |
| Tetrahydrofuran  | Haut<br>MAK-KZGW: 100 ppm<br>15 Minuten<br>MAK-KZGW: 300 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 50 ppm 8<br>Stunden<br>MAK-TMW: 150 mg/m <sup>3</sup><br>8 Stunden | TWA: 50 ppm 8 timer<br>TWA: 150 mg/m <sup>3</sup> 8 timer<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutter<br>STEL: 100 ppm 15<br>minutter<br>Hud | Haut/Peau<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 300 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 150 mg/m <sup>3</sup> 8<br>Stunden | STEL: 300 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 150 mg/m <sup>3</sup> 8<br>godzinach | TWA: 50 ppm 8 timer<br>TWA: 150 mg/m <sup>3</sup> 8 timer<br>STEL: 75 ppm 15<br>minutter. value<br>calculated<br>STEL: 187.5 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated<br>Hud |

|                  |                                                                                                                    |                                                                                                                                                                               |                                                                                                                                |                                                                                                                                         |                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Component</b> | <b>Bulgaria</b>                                                                                                    | <b>Croatia</b>                                                                                                                                                                | <b>Ireland</b>                                                                                                                 | <b>Cyprus</b>                                                                                                                           | <b>Czech Republic</b>                                                                                                |
| Tetrahydrofuran  | TWA: 50.0 ppm<br>TWA: 150.0 mg/m <sup>3</sup><br>STEL : 100 ppm<br>STEL : 300.0 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 50 ppm 8<br>satima.<br>TWA-GVI: 150 mg/m <sup>3</sup> 8<br>satima.<br>STEL-KGVI: 100 ppm<br>15 minutama.<br>STEL-KGVI: 300 mg/m <sup>3</sup><br>15 minutama. | TWA: 50 ppm 8 hr.<br>TWA: 150 mg/m <sup>3</sup> 8 hr.<br>STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15<br>min<br>Skin | Skin-potential for<br>cutaneous absorption<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 150 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 300 mg/m <sup>3</sup> |

|                  |                                                                                                                                                                   |                                                                                                                                       |                                                                                            |                                                                                                                                                                                                                 |                                                                                                                                                    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Component</b> | <b>Estonia</b>                                                                                                                                                    | <b>Gibraltar</b>                                                                                                                      | <b>Greece</b>                                                                              | <b>Hungary</b>                                                                                                                                                                                                  | <b>Iceland</b>                                                                                                                                     |
| Tetrahydrofuran  | Nahk<br>TWA: 50 ppm 8<br>tundides.<br>TWA: 150 mg/m <sup>3</sup> 8<br>tundides.<br>STEL: 100 ppm 15<br>minutites.<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutites. | Skin notation<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15<br>min | STEL: 250 ppm<br>STEL: 735 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | STEL: 300 mg/m <sup>3</sup> 15<br>percekben. CK<br>STEL: 100 ppm 15<br>percekben. CK<br>TWA: 150 mg/m <sup>3</sup> 8<br>órában. AK<br>TWA: 50 ppm 8 órában.<br>AK<br>lehetséges borón<br>keresztüli felszívódás | STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm 8<br>klukkustundum.<br>TWA: 150 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Skin notation |

|                  |                                                                                                                                         |                                                                                                            |                                                                                                                                                                                                          |                                                                                                                                                                              |                                                                                                                                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Component</b> | <b>Latvia</b>                                                                                                                           | <b>Lithuania</b>                                                                                           | <b>Luxembourg</b>                                                                                                                                                                                        | <b>Malta</b>                                                                                                                                                                 | <b>Romania</b>                                                                                                                                   |
| Tetrahydrofuran  | skin - potential for<br>cutaneous exposure<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 50 ppm IPRD<br>TWA: 150 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup> | Possibility of significant<br>uptake through the skin<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 150 mg/m <sup>3</sup> 8<br>Stunden<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 300 mg/m <sup>3</sup> 15<br>Minuten | possibility of significant<br>uptake through the skin<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup><br>STEL: 100 ppm 15<br>minuti<br>STEL: 300 mg/m <sup>3</sup> 15<br>minuti | Skin notation<br>TWA: 50 ppm 8 ore<br>TWA: 150 mg/m <sup>3</sup> 8 ore<br>STEL: 100 ppm 15<br>minute<br>STEL: 300 mg/m <sup>3</sup> 15<br>minute |

|                  |                            |                                                                                                                      |                                                                                                                                             |                                                                                                                                                                         |                                                                                                                                           |
|------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Component</b> | <b>Russia</b>              | <b>Slovak Republic</b>                                                                                               | <b>Slovenia</b>                                                                                                                             | <b>Sweden</b>                                                                                                                                                           | <b>Turkey</b>                                                                                                                             |
| Tetrahydrofuran  | MAC: 100 mg/m <sup>3</sup> | Ceiling: 300 mg/m <sup>3</sup><br>Potential for cutaneous<br>absorption<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 50 ppm 8 urah<br>TWA: 150 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 100 ppm 15<br>minutah<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutah | Binding STEL: 100 ppm<br>15 minuter<br>Binding STEL: 300<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 50 ppm 8 timmar.<br>NGV<br>TLV: 150 mg/m <sup>3</sup> 8<br>timmar. NGV | Deri<br>TWA: 50 ppm 8 saat<br>TWA: 150 mg/m <sup>3</sup> 8 saat<br>STEL: 100 ppm 15<br>dakika<br>STEL: 300 mg/m <sup>3</sup> 15<br>dakika |

## Biological limit values

List source(s):

|                  |                       |                       |               |                                               |                                                  |
|------------------|-----------------------|-----------------------|---------------|-----------------------------------------------|--------------------------------------------------|
| <b>Component</b> | <b>European Union</b> | <b>United Kingdom</b> | <b>France</b> | <b>Spain</b>                                  | <b>Germany</b>                                   |
| Tetrahydrofuran  |                       |                       |               | Tetrahydrofuran: 2 mg/L<br>urine end of shift | Tetrahydrofuran: 2 mg/L<br>urine (end of shift ) |

|                  |                  |               |                        |                   |               |
|------------------|------------------|---------------|------------------------|-------------------|---------------|
| <b>Component</b> | <b>Gibraltar</b> | <b>Latvia</b> | <b>Slovak Republic</b> | <b>Luxembourg</b> | <b>Turkey</b> |
|------------------|------------------|---------------|------------------------|-------------------|---------------|

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|                 |  |  |                                                                   |  |  |
|-----------------|--|--|-------------------------------------------------------------------|--|--|
| Tetrahydrofuran |  |  | Tetrahydrofuran: 2 mg/L<br>urine end of exposure or<br>work shift |  |  |
|-----------------|--|--|-------------------------------------------------------------------|--|--|

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                          | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) |                                 |                                    |                                   | DNEL = 12.6mg/kg<br>bw/day           |

| Component                          | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | DNEL = 300mg/m <sup>3</sup>         | DNEL = 96mg/m <sup>3</sup>             | DNEL = 150mg/m <sup>3</sup>           | DNEL = 72.4mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                          | Fresh water     | Fresh water<br>sediment         | Water Intermittent | Microorganisms in<br>sewage treatment | Soil (Agriculture)          |
|------------------------------------|-----------------|---------------------------------|--------------------|---------------------------------------|-----------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | PNEC = 4.32mg/L | PNEC = 23.3mg/kg<br>sediment dw | PNEC = 21.6mg/L    | PNEC = 4.6mg/L                        | PNEC = 2.13mg/kg<br>soil dw |

| Component                          | Marine water     | Marine water<br>sediment        | Marine water<br>Intermittent | Food chain             | Air |
|------------------------------------|------------------|---------------------------------|------------------------------|------------------------|-----|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | PNEC = 0.432mg/L | PNEC = 2.33mg/kg<br>sediment dw |                              | PNEC = 67mg/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material  | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|-----------------|-------------------|-----------------|-------------|-----------------------|
| Butyl rubber    | 10 minutes        | 0.3 mm          | EN 374      | (minimum requirement) |
| Neoprene gloves |                   |                 |             |                       |

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**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** In case of insufficient ventilation, wear suitable respiratory equipment  
**Recommended Filter type:** Multi-purpose/ABEK conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              |                          |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | Highly flammable         | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | -21 °C / -5.8 °F         | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Immiscible               |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Tetrahydrofuran                                | 0.45                     |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 0.878 g/cm3              | @ 20 °C                                  |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### 9.2. Other information

|                                                                         |                                                      |
|-------------------------------------------------------------------------|------------------------------------------------------|
| <b>Molecular Formula</b>                                                | C4 H11 BO                                            |
| <b>Molecular Weight</b>                                                 | 85.94                                                |
| <b>Explosive Properties</b>                                             | Vapors may form explosive mixtures with air          |
| <b>Substances/mixtures which, in contact with water, emit flammable</b> | Emitted gas ignites spontaneously Gas(es) = Hydrogen |

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gases

## Section 10: Stability and reactivity

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Air sensitive. Water reactive.

### 10.3. Possibility of hazardous reactions

**Hazardous Polymerization**  
**Hazardous Reactions**

No information available.  
None under normal processing. Reacts violently with water.

### 10.4. Conditions to avoid

Exposure to moist air or water. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Oxidizing agent.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Oxides of boron. Hydrogen.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component       | LD50 Oral          | LD50 Dermal           | LC50 Inhalation                               |
|-----------------|--------------------|-----------------------|-----------------------------------------------|
| Tetrahydrofuran | 1650 mg/kg ( Rat ) | > 2000 mg/kg (Rabbit) | 180 mg/L ( Rat ) 1 h<br>53.9 mg/L ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Category 2

#### (c) serious eye damage/irritation;

Category 1

#### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                          | Test method                                       | Test species | Study result    |
|------------------------------------|---------------------------------------------------|--------------|-----------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | Local Lymph Node Assay<br>OECD Test Guideline 429 | mouse        | non-sensitising |

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(e) germ cell mutagenicity; No data available

| Component                          | Test method                                             | Test species          | Study result |
|------------------------------------|---------------------------------------------------------|-----------------------|--------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | OECD Test Guideline 476<br>Gene cell mutation           | in vivo<br>Mammalian  | negative     |
|                                    | OECD Test Guideline 473<br>Chromosomal aberration assay | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; Category 2

Limited evidence of a carcinogenic effect The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component       | EU | UK | Germany | IARC     |
|-----------------|----|----|---------|----------|
| Tetrahydrofuran |    |    |         | Group 2B |

(g) reproductive toxicity; No data available

| Component                          | Test method             | Test species / Duration | Study result      |
|------------------------------------|-------------------------|-------------------------|-------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) | OECD Test Guideline 416 | Rat<br>2 Generation     | NOAEL = 3,000 ppm |

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system, Central nervous system (CNS).

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

Ecotoxicity effects

| Component       | Freshwater Fish                                                                        | Water Flea                                   | Freshwater Algae |
|-----------------|----------------------------------------------------------------------------------------|----------------------------------------------|------------------|
| Tetrahydrofuran | 2160 mg/l LC50 = 96 h<br>Pimephales promelas<br>Leuciscus idus: LC50: 2820<br>mg/L/48h | EC50 48 h 3485 mg/l<br>EC50: >10000 mg/L/24h |                  |

### 12.2. Persistence and degradability

Persistence Persistence is unlikely.

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## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Tetrahydrofuran | 0.45    | No data available             |

## 12.4. Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and floats on water Is not likely mobile in the environment due its low water solubility.

## 12.5. Results of PBT and vPvB assessment

No data available for assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

| Component       | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|-----------------|------------------------------------------|--------------------------------------------------|
| Tetrahydrofuran | Group III Chemical                       |                                                  |

## 12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## Section 14: Transport information

### IMDG/IMO

#### 14.1. UN number

UN3148

#### 14.2. UN proper shipping name

WATER-REACTIVE LIQUID, N.O.S.

#### Technical Shipping Name

(Borane-tetrahydrofuran complex)

#### 14.3. Transport hazard class(es)

4.3

#### 14.4. Packing group

I

### ADR

ALFAAR21730

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**14.1. UN number** UN3148  
**14.2. UN proper shipping name** WATER-REACTIVE LIQUID, N.O.S.  
**Technical Shipping Name** (Borane-tetrahydrofuran complex)  
**14.3. Transport hazard class(es)** 4.3  
**14.4. Packing group** I

## IATA

**14.1. UN number** UN3148  
**14.2. UN proper shipping name** WATER-REACTIVE LIQUID, N.O.S.\*  
**Technical Shipping Name** (Borane-tetrahydrofuran complex)  
**14.3. Transport hazard class(es)** 4.3  
**14.4. Packing group** I

**14.5. Environmental hazards** No hazards identified  
**14.6. Special precautions for user** No special precautions required.  
**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                 | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-------------------------------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Tetrahydrofuran                           | 109-99-9   | 203-726-8 | -      | -   | X     | X    | KE-33454 | X    | X    |
| Boron, trihydro(tetrahydrofuran)-, (T-4)- | 14044-65-6 | 237-881-8 | -      | -   | X     | X    | -        | -    | X    |

| Component                                 | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-------------------------------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Tetrahydrofuran                           | 109-99-9   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Boron, trihydro(tetrahydrofuran)-, (T-4)- | 14044-65-6 | X    | ACTIVE                                        | -   | X    | -    | X     | -     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component                                 | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran                           | 109-99-9   | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)           | -                                                                                                     |
| Boron, trihydro(tetrahydrofuran)-, (T-4)- | 14044-65-6 | -                                                                   | -                                                                             | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

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## Seveso III Directive (2012/18/EC)

| Component                                 | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Tetrahydrofuran                           | 109-99-9   | Not applicable                                                                            | Not applicable                                                                           |
| Boron, trihydro(tetrahydrofuran)-, (T-4)- | 14044-65-6 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 2 (self classification)

| Component                                 | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-------------------------------------------|---------------------------------------|-------------------------|
| Tetrahydrofuran                           | WGK1                                  |                         |
| Boron, trihydro(tetrahydrofuran)-, (T-4)- | WGK2                                  |                         |

| Component       | France - INRS (Tables of occupational diseases)      |
|-----------------|------------------------------------------------------|
| Tetrahydrofuran | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                          | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 90 ) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

## Full text of H-Statements referred to under sections 2 and 3

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H260 - In contact with water releases flammable gases which may ignite spontaneously  
H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
EUH014 - Reacts violently with water  
EUH019 - May form explosive peroxides  
H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Prepared By** Health, Safety and Environmental Department

**Revision Date** 07-Dec-2024

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

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## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**